

Randomized Controlled Trial of Intravenous Immunoglobulin Versus Oral Prednisolone in Chronic Inflammatory Demyelinating Polyradiculoneuropathy

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This multicenter, randomized, double-blind, crossover trial compared a six week course of oral prednisolone tapering from 60 mg to 10 mg daily with intravenous immunoglobulin (IVIg) 2.0 g/kg given over one to two days for treating chronic inflammatory demyelinating polyradiculoneuropathy (CIDP). Twenty-four of the thirty-two randomized patients completed both treatment periods. Both treatments produced significant improvements in the primary outcome measure, change in an 11-point disability scale two weeks after randomization. There was slightly, but not significantly, more improvement after IVIg than with prednisolone, the mean difference between the groups in change in disability grade being 0.16 (95% CI = -0.35 to 0.66). There were also slightly, but not significantly, greater improvements favoring IVIg in the secondary outcome measures: time to walk 10 meters after two weeks and improvement in disability grade after six weeks. Results may have been biased against IVIg by the eight patients who did not complete the second arm of the trial. A serious adverse event (psychosis) attributable to treatment occurred in one patient while on prednisolone and in none with IVIg.

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Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is an important treatable cause of prolonged disability affecting at least one or two per 100,000 of the population.^{1,2} Research criteria for its diagnosis require progressive or relapsing weakness attributable to a demyelinating polyradiculoneuropathy, strict neurophysiological criteria indicating slowing or block of nerve conduction, optional neuropathological criteria of demyelinating or demyelinated nerve fibers on nerve biopsy, and absence of alternative causes.³ In a population-based study, 13% of patients required aid to walk and 54% were still receiving treatment on the prevalence date.¹ Between 3% and 7% of patients have

died during follow-up in the largest published series.⁴⁻⁶

An autoimmune etiology for CIDP is inferred from its resemblance to Guillain-Barré syndrome and experimental autoimmune neuritis and from its response to immunomodulatory treatment. Short-term responses to steroids,⁷ plasma exchange (PE),^{8,9} and intravenous immunoglobulin (IVIg)^{10,11} have all been shown in randomized trials. Some cases appear to respond to immunomodulatory drugs, including azathioprine,¹² cyclophosphamide,¹³ cyclosporin,¹⁴⁻¹⁶ interferon-alpha,¹⁷ and interferon-beta.^{18,19} The efficacy of these treatments has not been established by randomized trials. A

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crossover trial compared PE with IVIg in 20 patients and showed no difference in efficacy.²⁰ No other comparative trials are known to have been undertaken. We conducted a double-blind, crossover trial to determine whether a standard IVIg regimen is more efficacious than a standard oral steroid regimen in CIDP.

Patients and Methods

Patients

Between July, 1998, and November, 1999, we randomized patients with CIDP, attending nine European centers, who fulfilled the inclusion and exclusion criteria described in Table 1 and the neurophysiological criteria in Table 2. Before randomization, patients underwent a thorough medical and neurological examination and hematological and biochemical testing, including serum protein electrophoresis. The ethics committees of all participating institutions had approved the trial protocol. Patients received oral and written information about the trial and signed a form of consent before being randomized.

Treatment Allocation

Patients were randomly allocated to the treatment sequences, prednisolone followed by IVIg or IVIg followed by prednisolone (Fig). The sequences were stratified by center with a block size of 2 according to a sequence of random numbers provided by the trial statistician to the manufacturers who prepared the randomization packs. The block size was not revealed to the trialists. Each center was provided with coded packs containing either Sandoglobulin or placebo and tablets containing either placebo or prednisolone. Novartis prepared the coded packs from a random sequence generated by the trial statistician that was not revealed to the investigators except in an emergency. The IVIg infusions were given as 1.0 g/kg on 2 consecutive days or 2.0 g/kg in 24 hours. Placebo infusions contained albumin 6 mg/100 ml and were identical in appearance to the IVIg infusions. The prednisolone regimen was 60 mg each morning for 2 weeks, 40 mg each morning for 1 week, 30 mg each morning for 1 week, 20 mg each morning for 1 week, and then 10 mg each morning for 1 week. The intention was to wait for 4 weeks after completing the first (washout period) before starting the second treatment period. If a patient improved so much during the first treatment period that he or she did not need treatment as judged by the treating neurologist, the second period was delayed until treatment was necessary. If the patient worsened after starting the first treatment period, the treating neurologist might transfer the patient to the second period provided that at least 2 weeks had elapsed after the start of the first period. Worsening had to be accompanied by objective signs and an increase in either the lower limb or the upper limb disability by 1 point. If the patient worsened by a similar amount after 2 weeks from start of the second period, the treatment code might be broken and any treatment deemed appropriate given.

Assessments

A neurologist who did not have access to laboratory data or the liberty to ask questions about adverse events assessed pa-

Table 1. Inclusion and Exclusion Criteria

Inclusion criteria

- Diagnosis of CIDP by a consultant neurologist
- Progressive or relapsing motor and sensory dysfunction of more than one limb resulting from neuropathy developing over >2 months
- Reduced or absent tendon reflexes
- CSF less than 10 white cells/ μ l (CSF examination encouraged but not mandatory)
- Fulfilment of INCAT neurophysiological criteria for multifocal demyelinating polyradiculoneuropathy (see Table 2)
- Significant disability in upper or lower limb function (at least arm disability grade 2 or leg disability grade 1)
- Stable or worsening clinical condition (not improving spontaneously)

Exclusion criteria

- Associated systemic diseases that might cause neuropathy, such as carcinoma, systemic lupus erythematosus, diabetes mellitus, paraproteinemia (detectable by conventional high-resolution agarose gel electrophoresis), or HIV infection
- Pregnancy or planned pregnancy
- Severe concurrent medical conditions or previous neurological deficit that would interfere with treatment, including active peptic ulceration, renal failure, known requirement for steroids, or known allergic reaction to IVIg
- Significant respiratory impairment
- Treatment with steroids, IVIg, plasma exchange, or any immunosuppressant drug during the 6 weeks before randomization, except that treatment with azathioprine will be permitted provided that the dose has been unchanged for 6 months and is expected to remain unchanged
- Age less than 18 years
- Pure motor syndrome fulfilling criteria for multifocal motor neuropathy with conduction block
- Previous failure to respond to either IVIg or prednisolone

tients at entry and after 2, 4, and 6 weeks into each treatment period. Assessments included a novel disability scale (Table 3). A very similar scale has been validated as reproducible and responsive to change.³¹ Other assessments included timed 10-meter walk, 9-hole pegboard test (average of two sides),²¹ MRC sum score,²² maximum grip strength, Rotterdam handicap scale,³¹ modified Rankin scale,²³ sensory sum score,²⁴ and the short form 36 (SF-36) quality-of-life scale. Motor nerve conduction was measured in the right median and ulnar nerves at the start and end of each treatment period. The sum of the distal compound muscle action potential amplitudes and average of the maximal forearm conduction velocities were analyzed. A different neurologist applied a questionnaire about adverse events after 2, 4, and 6 weeks.

Statistical Analysis

The primary analysis compared the differences between treatment A and treatment B according to the intention-to-treat principle and using the methods described by Senn²⁵ to analyze the effect of treatment order. The major outcome cri-

Table 2. INCAT Neurophysiological Criteria for Diagnosis of CIDP

Either

1. Partial conduction block or abnormal temporal dispersion of conduction must be present in at least two nerves, and there must be significantly reduced conduction velocity, or significantly prolonged distal motor latency, or absent or significantly prolonged minimum F wave latency in at least one other nerve

or

2. In the absence of block or dispersion, significantly reduced conduction velocity, or significantly prolonged distal motor latency, or absent or significantly prolonged minimum F wave latency must be present in at least three nerves

or

3. In the presence of significant neurophysiological abnormalities in only two nerves, unequivocal histological evidence of demyelinating or demyelinated nerve fibers in a nerve biopsy must also be present

Recording technique

The following nerves were tested on both sides (unless the criteria are fulfilled by studying a smaller number of nerves or points)

Median (wrist, elbow, axilla)

Ulnar (wrist, elbow, axilla, Erb's point)

Peroneal (ankle, below fibular head, above fibular head)

Tibial (ankle, popliteal fossa)

Ten consecutive F waves were recorded from each nerve, and the minimal latency was measured

Definitions

Partial conduction block = <15% change in duration between proximal and distal sites and >20% drop in negative peak area and peak to peak amplitude

Abnormal temporal dispersion and possible conduction block = >15% change in duration between proximal and distal sites and >20% drop in negative peak area or peak to peak amplitude between proximal and distal sites; additional studies such as stimulation across short segments or recording of individual motor units required for confirmation

Significantly reduced conduction velocity = <80% of lower limit of normal or, if distal motor compound muscle action potential amplitude <80% of normal, <70% of lower limit of normal

Significantly prolonged distal motor latency = >125% of upper limit of normal or, if amplitude <80% of normal, >150% of upper limit of normal

Significantly delayed minimum F wave latency = >120% of upper limit of normal or, if amplitude <80% of normal, >150% of upper limit of normal

These criteria have been compared to those of the Ad Hoc Subcommittee³ and found to be more sensitive.

terion was the difference in the overall disability scale after 2 weeks. Secondary outcome criteria were changes in the 10-meter walk time and 9-hole pegboard time after 2 weeks and an assessment of each measure taking into account all time points. The effect on disability grade during the complete 6

week treatment period was calculated by computing the area under the curve described by the plots of disability grade against time above the disability grade at the start of that treatment period. Missing values at 4 or 6 weeks were treated as being the same as the last available value. We intended to randomize as many patients as possible within 3 years, expecting to accrue 40 patients. This would have been sufficient to show as significant at the 5% level with 80% power an effect whose magnitude was of the same order as that of 1 standard deviation in the change of disability grade. It would also have been one-third more than was necessary to detect a difference in efficacy between placebo and IVIg in a trial that had just been completed.²⁶

Results

The start of the trial was delayed for administrative reasons and recruitment had to be terminated after 32 patients had been randomized because the trial medication would have exceeded its expiry date. Twenty-four of the 32 patients randomized completed both treatment periods (Table 4). Eight patients were withdrawn before reaching the 2-week outcome measure in the second treatment period. In 6, the first arm was IVIg. In 3 of them (2 following improvement by 1 or more disability grades and 1 despite worsening by 1 grade), further treatment was not deemed necessary, so the second treatment was not started. One other patient had improved 2 and 4 weeks after receiving IVIg in the first treatment period, worsened, was withdrawn, received open IVIg, and improved again. She did not enter the second treatment period. The fifth patient improved by 2 disability grades following IVIg and then worsened by 4 grades after 1 week on prednisolone and was withdrawn to receive open IVIg. The sixth patient was withdrawn after 4 weeks, when a small cell lung carcinoma was discovered. Two of the

Fig. Diagram of patient randomization and progress.

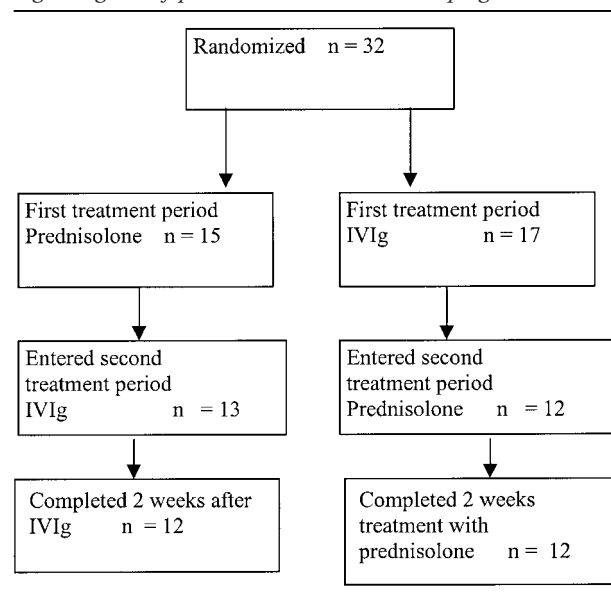


Table 3. INCAT Disability Scale

Arm disability	
0	No upper limb problems
1	Symptoms, in one or both arms, not affecting the ability to perform any of the following functions: doing all zips <i>and</i> buttons; washing <i>or</i> brushing hair; using a knife and fork together; and handling small coins
2	Symptoms, in one arm or both arms, affecting but not preventing any of the above-mentioned functions
3	Symptoms, in one arm or both arms, preventing one or two of the above-mentioned functions
4	Symptoms, in one arm or both arms, preventing three or all of the functions listed, but some purposeful movements still possible
5	Inability to use either arm for any purposeful movement
Leg disability	
0	Walking not affected
1	Walking affected, but walks independently outdoors
2	Usually uses unilateral support (stick, single crutch, one arm) to walk outdoors
3	Usually uses bilateral support (sticks, crutches, frame, two arms) to walk outdoors
4	Usually uses wheelchair to travel outdoors, but able to stand and walk a few steps with help
5	Restricted to wheelchair, unable to stand and walk a few steps with help
Overall disability = Sum of arm and leg disability	

patients who did not receive the second treatment had received prednisolone in the first treatment period. One was withdrawn because of the development of psychosis. The disability grade of the other was unchanged after the 6-week course of prednisolone, but the patient preferred not to enter the second treatment period.

In the primary analysis of the 24 patients who completed both treatment periods, the disability grade improved after 2 weeks by a mean (SD) of 0.58 (0.93) grades in the prednisolone treatment period ($p = 0.005$) and by 0.71 (1.27) grades in the IVIg treatment

period ($p = 0.012$). The mean difference between the treatment effects in the 2 periods was not significant, being 0.16 grades [95% confidence limits (CL) = -0.35 to 0.66] more improvement after IVIg than during prednisolone. A two-group t test showed no evidence of a carryover effect.

There were similar improvements after the prednisolone and IVIg treatment periods in the secondary outcome measures (Table 5). Although it was not pre-specified as a secondary outcome measure, we also determined the change in the primary outcome measure after 6 weeks. The improvement in disability grade in the prednisolone period compared to baseline was 0.62 (1.53), and the equivalent improvement with IVIg was 0.71 (1.19). Tests for carryover effects on these analyses were not significant. None of these outcome measures revealed significant differences between the two treatments. We also conducted analyses of the change in MRC sum score, grip strength, sensory sum score, Rotterdam handicap scale, and modified Rankin scale using the area under the curve method specified in the protocol, and none of these showed significant differences between the treatments. We found significant improvements in MRC sum score, grip strength, and sensory sum score with each treatment after 2 weeks and in MRC sum score and grip strength with IVIg and in sensory sum score with both treatments after 6 weeks. However, the differences in effect size were small, and, apart from greater improvement in grip strength after 6 weeks with IVIg, there were no significant differences between the treatments in the amount of improvement at either time. There were no significant ($p < 0.05$) differences in the neurophysiological measures, modified Rankin scale, Rotterdam handicap scale, or SF-36 physical functioning score after 6 weeks.

The analysis of the primary outcome measure was also undertaken on the first arm of the trial, including

Table 4. Clinical Features of Trial Participants

	All Participants	Participants with 2-Week Outcome for Both Treatment Periods	Participants Randomized to Prednisolone First	Participants Randomized to IVIg First
n	32	24	15	17
M/F	21/11	15/9	9/6	12/5
Age (years)	54.1 (17.1)	52.3 (17.6)	52.1 (18.3)	55.8 (16.2)
RR/progressive	20/12	16/8	10/5	10/7
Mean duration (years)	5.2 (7.1)	6.5 (7.8)	5.2 (6.5)	5.3 (7.8)
Worst arm disability grade in any attack	2.4 (1.3)	2.2 (1.3)	2.2 (1.1)	2.6 (1.4)
Worst leg disability grade in any attack	2.1 (1.5)	1.8 (1.3)	1.6 (0.8)	2.5 (1.8)
Disability grade at randomization	3.81 (1.71)	3.75 (1.75)	3.47 (1.30)	4.11 (2.00)

Numbers in parentheses are SD.

Table 5. Secondary Outcome Measures

	n	Prednisolone Mean (SD)	IVIg Mean (SD)	Difference Mean (SE) Prednisolone–IVIg	95% CL	<i>p</i>
Change in speed of walking over 10 m after 2 weeks (m/second)	24	0.01 (0.02)	0.01 (0.01)	0.00 (0.02)	−0.01 to 0.01	0.75
Change in time to complete 9-hole pegboard task after 2 weeks (seconds)	24	−2.40 (6.45)	−1.64 (3.87)	−0.76 (1.45)	−3.76 to 2.24	0.70
Disability grade area under the curve over 6 weeks (grade/time units)	24	−3.88 (5.64)	−3.13 (5.50)	−0.64 (1.34)	−3.42 to 2.15	0.64
Disability grade after 6 weeks	21	−0.62 (1.52)	−0.71 (1.19)	0.22 (0.40)	−0.62 to 1.06	0.59

all the patients who were randomized. The mean (SD) improvement after 2 weeks of the 15 patients treated with prednisolone first was 0.53 (0.74) grades ($p = 0.015$), and that of the 17 patients treated with IVIg first was 1.24 (1.75) grades ($p = 0.01$). There was 0.70 grades more improvement with IVIg than with prednisolone (95% CI = −0.29 to 1.70; $p = 0.16$).

Treatment, once started, was completed according to protocol in 29 of 30 IVIg courses and 24 of 27 prednisolone courses. Adverse events occurred after 18 of 30 IVIg courses and during 11 of 27 prednisolone courses. Headache developed in 10 (33%) of IVIg and 7 (26%) of prednisolone courses. Indigestion developed in 6 (20%) of IVIg and 4 (20%) of prednisolone courses. Fever developed in 5 (17%), rash in 2 (6%), and hypotension in 1 (3%) of IVIg courses but not in any prednisolone courses. None of these adverse events was statistically more common with either treatment. Urticaria led to the discontinuation of 1 IVIg course and psychosis to the discontinuation of 1 prednisolone course. Serious adverse events occurred in 3 patients. During the first treatment period, which utilized prednisolone, 1 patient was discovered to have a previously unrecognized carcinoma of the bronchus and was withdrawn from the trial. This patient did not receive IVIg and died 22 weeks after randomization. One patient developed psychosis during the first (prednisolone) treatment period and was withdrawn. This was the only serious adverse event attributed to treatment. The third patient developed heart failure requiring hospitalization 4 weeks after receiving IVIg in the first treatment period.

Discussion

Both oral prednisolone 60 mg daily for two weeks and IVIg 2.0 g/kg over one to two days significantly reduced disability and impairment in CIDP two weeks after onset of treatment. This result adds to the evidence from other randomized controlled trials that oral steroids⁷ and IVIg^{10,11,26,27} are beneficial. The improvement was slightly, but not significantly, greater

following IVIg than following prednisolone in 24 patients who completed at least two weeks of both treatment periods.

The trial was not designed and powered to detect equivalence between oral prednisolone and IVIg. The results allow only the statement that the difference in the overall improvement after two weeks of treatment will lay between 0.35 of a disability grade more improvement with prednisolone to 0.66 of a disability grade more improvement with IVIg in 95% of trials. This implies that a larger trial might have detected a relative benefit of IVIg over prednisolone of as much as 0.66 of a grade after two weeks. Because the scale grades are coarse, such a difference of 0.5 or more of a grade would probably seem worthwhile to patients.

The withdrawal of eight patients might have prevented the detection of a more significant difference. The analysis of the primary outcome measure was repeated excluding the patient who was found to have a carcinoma and using the same value at week two as had been reached at week 1 for one patient who entered the second (prednisolone) treatment period, deteriorated after one week, and was withdrawn. In addition, values of no change for the second treatment period were used for the patients who did not undergo the second treatment. With these assumptions, the mean (SD) improvement in disability grade after two weeks during the prednisolone treatment period was 0.29 (1.19), which was not significant ($p = 0.18$), and with IVIg 0.87 (1.46), which was significant ($p = 0.002$). There was a marked trend towards more improvement with IVIg than with prednisolone; the difference being 0.56 (95% CI = −0.05 to 1.17) of a grade more improvement in the IVIg group ($p = 0.07$).

With the exception of the development of psychosis attributed to prednisolone in one patient, the adverse events attributable to both treatments encountered in this short-term trial were minor and rarely resulted in interruption of treatment. One patient who had improved with IVIg deteriorated markedly within two

weeks after starting prednisolone, and such deterioration with predominantly motor CIDP has been documented previously.²⁸ The major concerns about both treatments are long-term effects. Many patients treated with steroids require treatment for many months, for years, or even for life. The side effects of such a regimen are well-known and include moon facies, obesity, osteoporosis, hypertension, diabetes, cataract, and mood changes and their consequences.^{29,30} The short-term costs of IVIg may be balanced by the longer term health care costs and disability caused by these long-term steroid side effects. IVIg is a human blood product and carries the theoretical risk of transmitting known and unknown infectious agents. Fortunately, the safety record of most immunoglobulin preparations, including that used in this study, is excellent, so this concern remains theoretical.

The results of this trial confirm the efficacy of both oral steroids and IVIg in CIDP. They suggest slightly more short-term efficacy but slightly more short-term adverse events with IVIg than with steroids. Some patients respond to their first course of treatment and do not require prolonged medication. For them, the choice of treatment is evenly balanced. For patients who require long-term treatment, IVIg may be preferable because of the known serious side effects of prolonged steroid treatment, but this requires confirmation in longer term trials. In their absence, because it is not possible to predict in advance who will respond to one treatment or the other, IVIg is likely to be selected as the initial treatment of choice.

Appendix

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